





1

A large, solid green circle is centered on a white background. Inside the circle, the number '2' is written in a white, bold, sans-serif font. The number is positioned in the center of the circle, both horizontally and vertically.

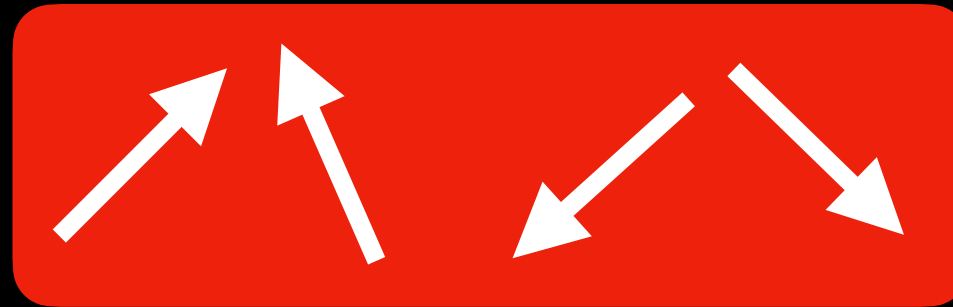
2



3

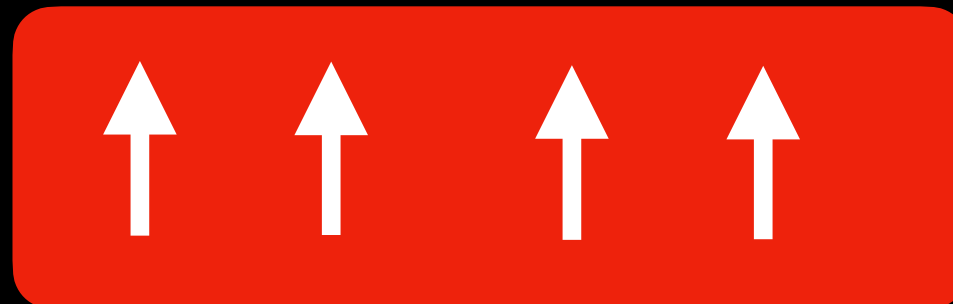
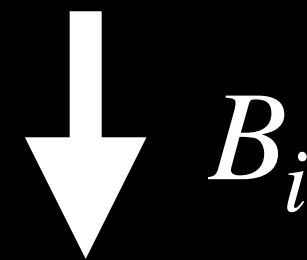
Step 3. Adiabatic demagnetization

1 Random spin, High Entropy



S_{spin} is large, $T = T_i$

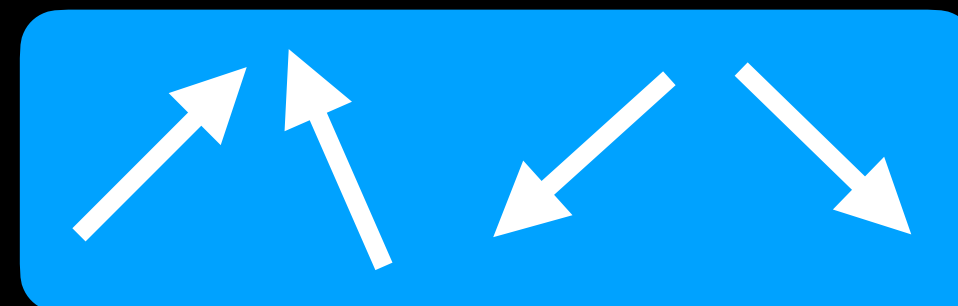
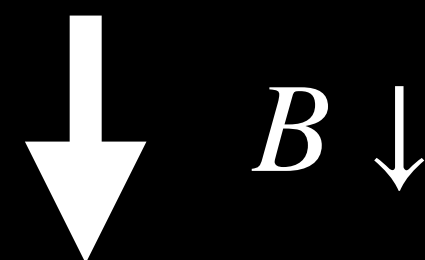
2 Apply magnetic field B
in thermal contact with the
helium bath ($T = 1.2$ K)



$E = -\mu \cdot \mathbf{B}$, $S_{\text{spin}} \downarrow$

Heat flows to the bath: $T = T_i$

3 Thermally isolate and
slow demagnetization



$B \downarrow$, $S_{\text{spin}} \uparrow$, $T \downarrow$

$B_i \rightarrow B_f$, $B_f \ll B_i$

$$T_f \approx T_i \frac{B_f}{B_i} \approx 10^{-2} \text{ K}$$

Step 4. Spin alignment